Precision and Recall

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Precision and Recall

> Prepare Precision and Recall: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Choose the **denominator** that answers the question

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Accuracy and Imbalance

Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4.

It **miss**es all four blocked cases, so the overall score hides complete failure on that evaluated class.

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Interpret calibration using groups of forecasts

Reference example, not the only acceptable wording: Teacher explanation: 2/10=.2 and 6/10=.6.

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

Concept to explain

Precision measures how many positive **prediction**s are right; **recall** measures how many actual positives are found.

Term	Meaning
precision	TP divided by TP plus FP
recall	TP divided by TP plus FN
false alarm	false-positive prediction
miss	false-negative prediction

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: **recall** 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **precision recall** and **denominator**s. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

The common class can dominate the total; inspect the full confusion matrix and class counts.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

End goal for the pictorial case

By the end, I can explain **precision** and **recall** using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Precision and Recall using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Calculate both and choose a priority for two consequence scenarios.

MODEL IDEA **Precision** measures how many positive **prediction**s are right; **recall** measures how many actual positives are found.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Confusion counts: TP=8, FP=2, FN=4, TN=16. **Precision** = 8 ÷ (8+2) = **80%**.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Confusion counts: TP=8, FP=2, FN=4, TN=16.

Step 2

STEP 2 **Precision** = 8 ÷ (8+2) = **80%**.

Step 3

STEP 3 **Recall** = 8 ÷ (8+4) ≈ 67%.

Step 4

STEP 4 Choose threshold and human review by comparing the cost of two **false alarm**s with four **miss**es.

CHECK Verify the **denominator**, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Positive is spotted: TP=8, FP=2, FN=4.

The teaching case

Compute **precision** and **recall**.

Exact explanation to model

Why the result follows

Precision=8/10=**80%**; **recall**=8/12 about 66.7%. **Denominator**s are predicted positives and actual positives respectively.

Explain the added worked example

Explain the added worked example · 1

Blocked remains positive. **Precision** asks: of the cases flagged blocked, how many were actually blocked according to the reference? **Recall** asks: of all reference-blocked cases, how many were found?

Both use true positives, but they answer different questions. Keep reference rows and **prediction** columns ordered clear then blocked.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **precision recall** and **denominator**s.

For matrix [[12, 4],[1, 3]], TN=12, FP=4, FN=1, TP=3. There are seven predicted positives and four reference positives. **Precision** is 3/7, about 42.9 percent. **Recall** is 3/4=75 percent. Accuracy is 15/20=75 percent.

A correct **denominator** matters more than memorizing a percentage.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Key terms

Term	Meaning
Precision	True positives divided by all predicted positives
Recall	True positives divided by all reference positives
Denominator	The group counted in the bottom of a fraction
Undefined	A ratio with no valid value because its denominator is zero

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Precision recall and denominators at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Precision	3 correct flags	7 total flags
Recall	3 found blocked cases	4 reference blocked
Accuracy	15 correct labels	20 total records

Draw a box around the predicted-blocked column for precision and around the reference-blocked row for recall. Their overlap is TP. A model can flag nearly everything and find most blocked cases while producing many false alarms. Conversely, a model can flag very few cases accurately while missing many positives.

Calculate precision, recall and accuracy for the worked matrix. Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Precision and Recall

Put **precision** and **recall denominator**s on the picture instead of memorising names.

The facts we will use

Supplied facts

Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/17

Keep precision and recall denominators distinct

	predicted spotted	predicted healthy
actual spotted	6	4
actual healthy	2	8

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/17

Precision and Recall

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predicted positives

Eight leaves were flagged: precision=6/(6+2)=75%.



2. Actual positives

Ten leaves were spotted: recall=6/(6+4)=60%.



3. Keep goals separate

Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

CHANGE ONE THING

Correct two false negatives into true positives.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Predicted positives	Eight leaves were flagged: precision=6/(6+2)=75%.
2. Actual positives	Ten leaves were spotted: recall=6/(6+4)=60%.
3. Keep goals separate	Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

Change one thing in the pictured case

Change one thing

Change: Correct two false negatives into true positives.

Predicted result

Prediction: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.

Why the result follows

Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Precision and Recall, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Positive is spotted: TP=8, FP=2, FN=4.

2. State the decision to explain

Write the question to explain: Compute **precision** and **recall**.

3. Reveal the worked reasoning

Reveal after **prediction**s: **Precision**=8/10=**80%**; **recall**=8/12 about 66.7%. **Denominator**s are predicted positives and actual positives respectively.

Hold back the independent task

Keep for an independent attempt: With disease positive, TP=6, FP=2 and FN=4. Calculate **precision** and **recall**, showing both **denominator**s.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and **denominator**s for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: **Precision** 3/7 about 42.9 percent, **recall** 3/4=75 percent, accuracy 15/20=75 percent. Exit response: Request TP, FP, FN, sample size, positive class and **precision**.

Finding every positive can coexist with many **false alarm**s.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Why can 80 percent accuracy hide missed blocked cases? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify precision in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: identify the groups; calculate both ratios. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between precision and recall. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Explain precision and recall for [[16, 0],[4, 0]].

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Choose the denominator that answers the question; Calculate precision and recall with a declared positive class. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion practice

Completion task: Calculate **precision**, **recall** and accuracy for the worked matrix.

Reasoning to listen for

Teacher check: Teacher explanation: **Precision** 3/7 about 42.9 percent, **recall** 3/4=75

percent, accuracy 15/20=75 percent.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Precision recall** and **denominator**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Predicted positives TP+FP=7; reference positives TP+FN=4.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No. **Precision** used8 predicted-positive leaves; **recall** used10 actual-positive leaves. Their meanings remain distinct even when numbers later coincide. Changed-case check: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.

TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Changed-case reasoning: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Collect each learner's explanation before discussion. Use the first **miss**ing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Precision and Recall investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Metric Tradeoff Lab: compute both metrics for two thresholds, map errors to consequences, and defend a choice.

3. Run one normal, one boundary, one **miss**ing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or **miss**ing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR **Precision** and **recall** are two names for the same accuracy calculation.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Predicted positives TP+FP=7; reference positives TP+FN=4.

A stronger claim makes a stronger project: Request TP, FP, FN, sample size, positive class and **precision**. Finding every positive can coexist with many **false alarm**s.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: TP=6, FP=2, FN=3. **Precision** 6/8=75 percent; **recall** 6/9 about 66.7 percent; accuracy 14/19 about 73.7 percent.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Precision and Recall questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Precision=8/10=**80%**; **recall**=8/12 about 66.7%. **Denominator**s are predicted positives and actual positives respectively.

Independent question

Independent question · withhold the key

With disease positive, TP=6, FP=2 and FN=4. Calculate **precision** and **recall**, showing both **denominator**s.

Expected independent answer

Teacher key · Expected independent answer

Precision is 6/(6+2) = 75%. **Recall** is 6/(6+4) = **60%**. **Precision** counts predicted positives; **recall** counts actual positives.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: "**Precision** and **recall** use the same **denominator**." Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Precision=9/12=75%; **recall**=9/15=**60%**. With no predicted positives, TP+FP=0 so **precision** is **undefined** mathematically; reporting software may apply a convention that must be disclosed.

Do not interpret a zero-**denominator** convention as observed performance.

Explanations for the additional workbook examples

W1 Identify the groups

Workbook question · W1

For the worked matrix, give predicted-positive and reference-positive totals.

Teacher reasoning · W1

Teacher explanation: Predicted positives TP+FP=7; reference positives TP+FN=4.

Teacher reasoning · W1

Predicted positives include true positives and false positives, so add TP + FP to get 7. Reference positives include true positives and false negatives, so add TP + FN to get 4.

These are different groups: one is defined by the **prediction**, the other by the reference label.

W2 Calculate both ratios

Workbook question · W2

Calculate **precision**, **recall** and accuracy for the worked matrix.

Teacher reasoning · W2

Teacher explanation: **Precision** 3/7 about 42.9 percent, **recall** 3/4=75 percent, accuracy 15/20=75 percent.

W3 Handle no flags

Workbook question · W3

Explain **precision** and **recall** for [[16, 0],[4, 0]].

Teacher reasoning · W3

Teacher explanation: **Precision** is **undefined**, with no flags; **recall** is 0/4=0. This does not mean excellent detection.

W4 Apply independently

Workbook question · W4

Use [[8, 2],[3, 6]]. Give TP, FP, FN, **precision**, **recall** and accuracy.

Teacher reasoning · W4

Teacher explanation: TP=6, FP=2, FN=3. **Precision** 6/8=75 percent; **recall** 6/9 about 66.7 percent; accuracy 14/19 about 73.7 percent.

W5 Limit a claim

Workbook question · W5

A team reports **recall** 100 percent but hides **precision** and counts. What is **miss**ing?

Teacher reasoning · W5

Teacher explanation: Request TP, FP, FN, sample size, positive class and **precision**. Finding every positive can coexist with many **false alarm**s.

Assess independent transfer

Independent question · withhold the key

Selected case: Use [[8, 2],[3, 6]]. Give TP, FP, FN, **precision**, **recall** and accuracy.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: TP=6, FP=2, FN=3. **Precision** 6/8=75 percent; **recall** 6/9 about 66.7 percent; accuracy 14/19 about 73.7 percent.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Precision and Recall to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with **denominator**s.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 18: Error Analysis. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **precision** and **recall** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: TP=6, FP=2, FN=3. **Precision** 6/8=75 percent; **recall** 6/9 about 66.7 percent; accuracy 14/19 about 73.7 percent.

Fresh retry [[5, 1],[2, 4]]: **precision** 4/5=80 percent, **recall** 4/6 about 66.7 percent, accuracy 9/12=75 percent. Require both **denominator**s. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Thresholds and consequences. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/18, ai-4/20, ai-4/23. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Precision and Recall lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Choose the **denominator** that answers the question

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Calculate **precision**, **recall** and accuracy for the worked matrix.

Supported practice answer

Reasoned practice answer: Teacher explanation: **Precision** 3/7 about 42.9 percent, **recall** 3/4=75 percent, accuracy 15/20=75 percent.

Check every learner before discussion: Use the worked case for **Precision recall** and **denominator**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Predicted positives TP+FP=7; reference positives TP+FN=4.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Correct two false negatives into true positives. Require a **prediction** and a reason before comparing. Expected result: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.

Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific **miss**ing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

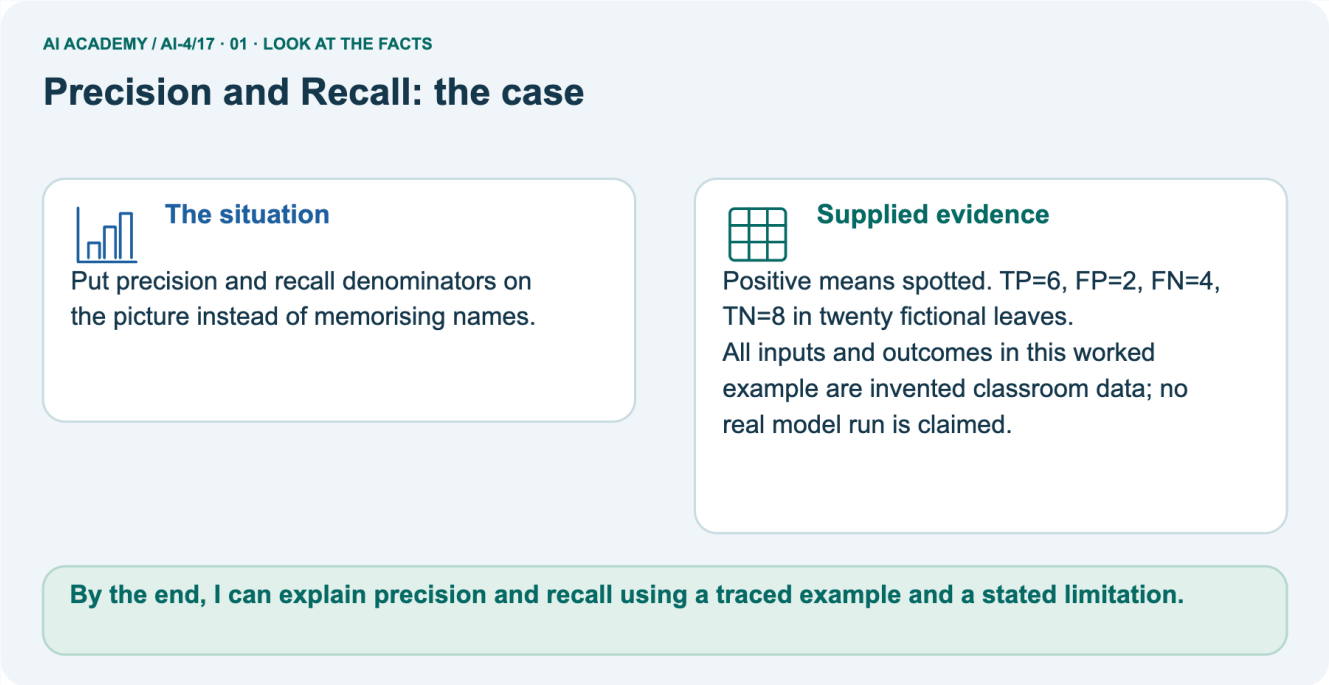
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Precision and Recall graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Precision and Recall: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Put **precision** and **recall denominator**s on the picture instead of memorising names. Supplied facts: Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

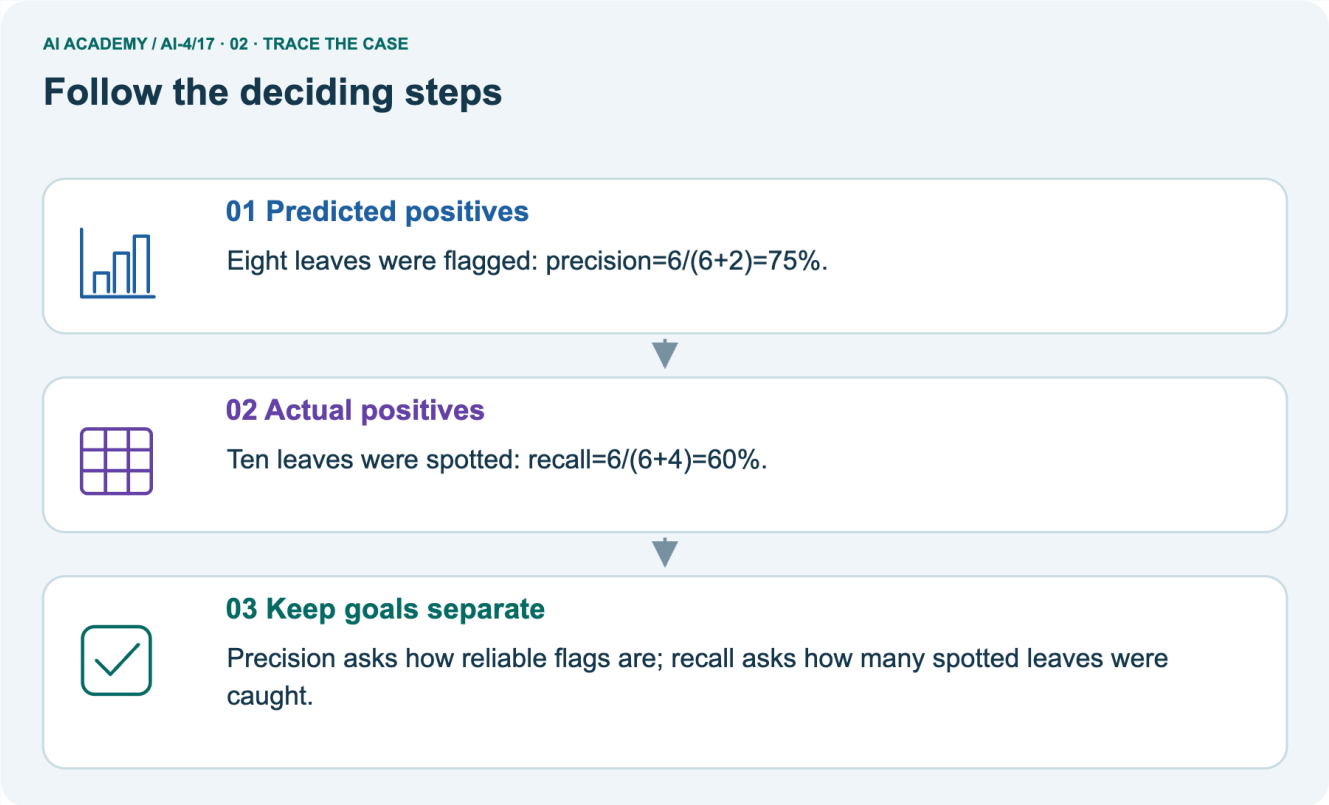
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

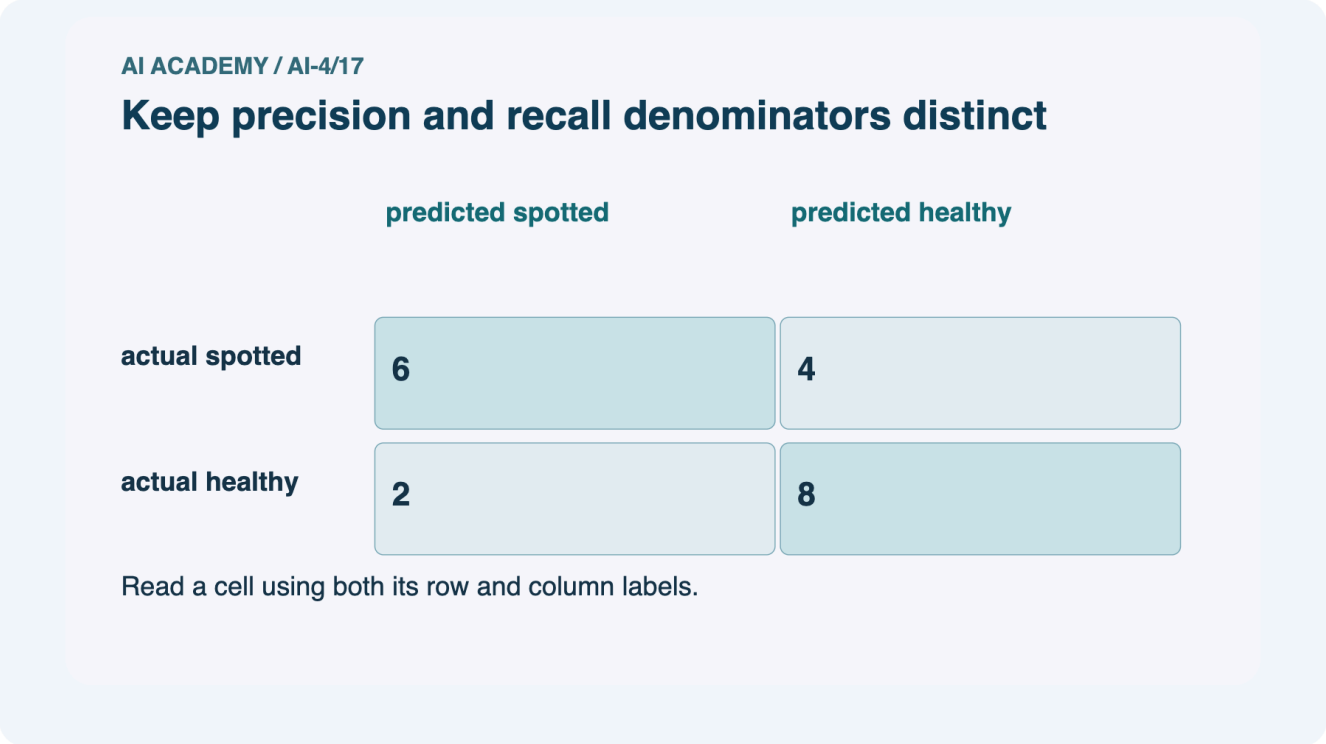
Predicted positives: Eight leaves were flagged: **precision**=6/(6+2)=75%. → Actual positives: Ten leaves were spotted: **recall**=6/(6+4)=**60%**. → Keep goals separate: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep precision and recall denominators distinct



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **precision** and **recall denominator**s distinct. actual spotted: predicted spotted = 6, predicted healthy = 4. actual healthy: predicted spotted = 2, predicted healthy = 8.

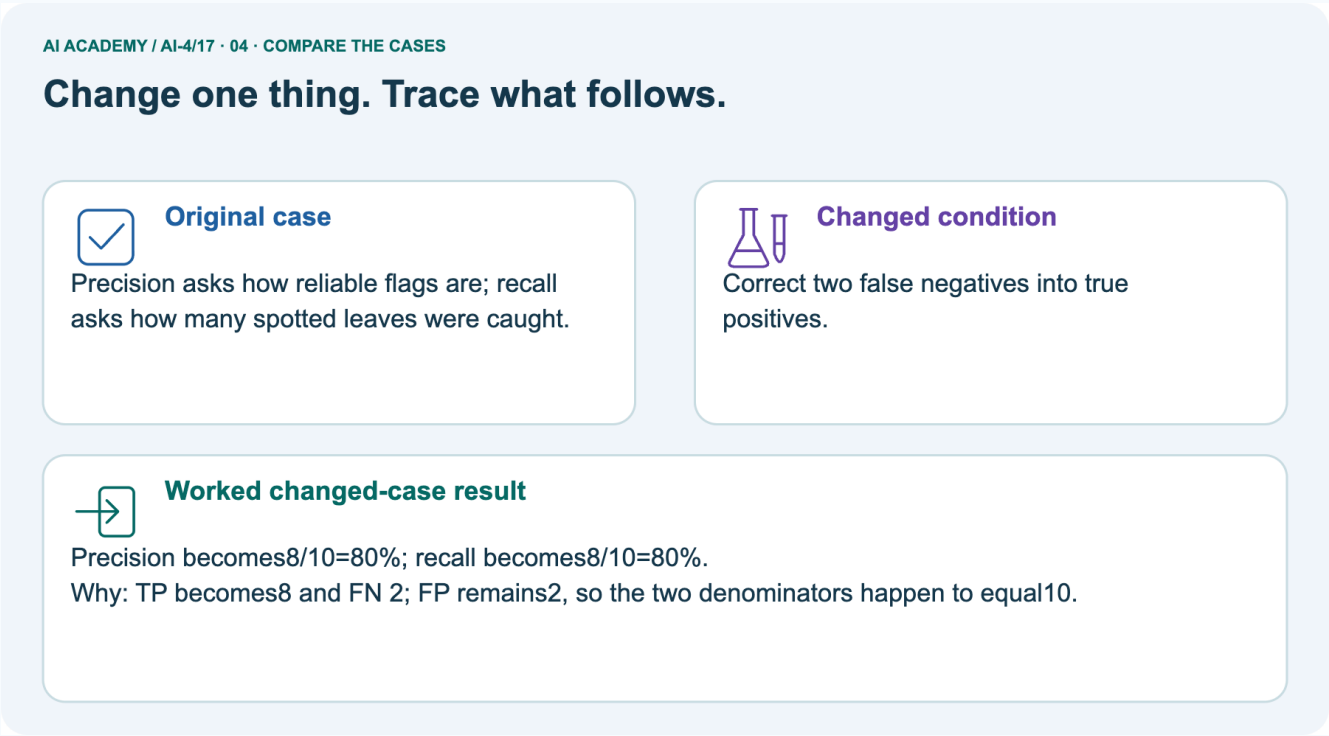
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and **denominator**s before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Changed condition: Correct two false negatives into true positives.

Expected result: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

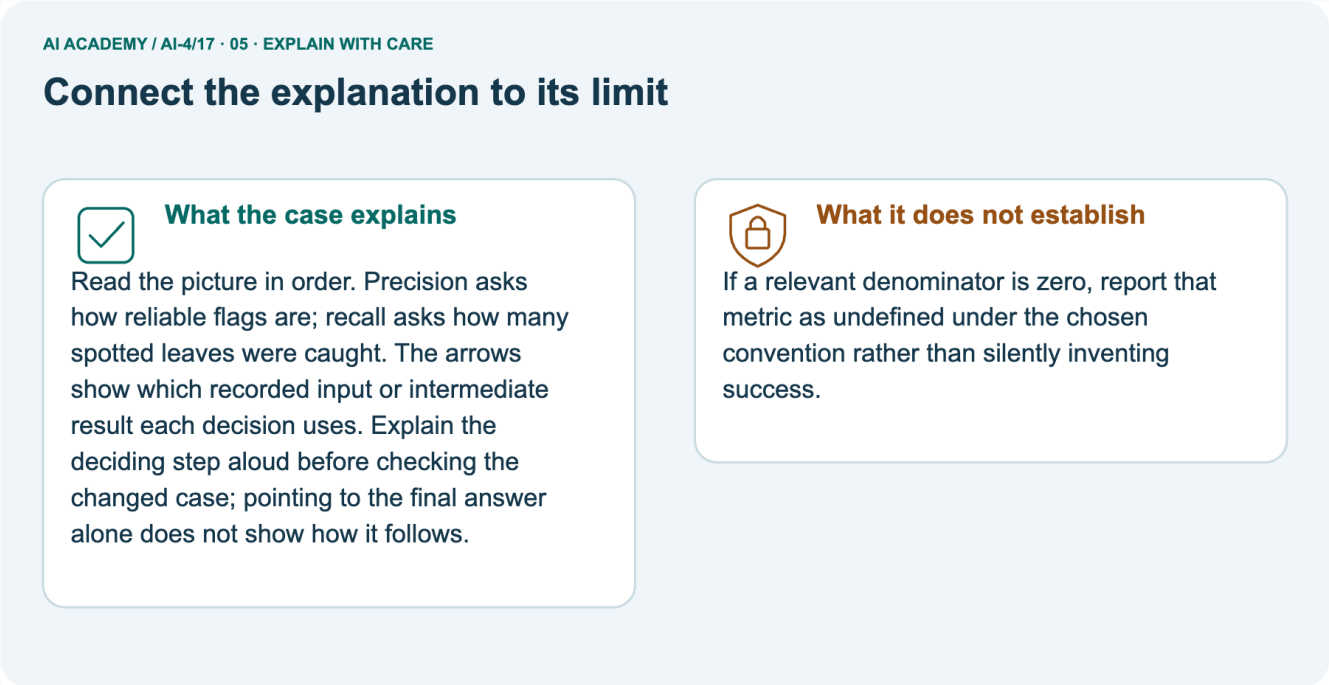
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

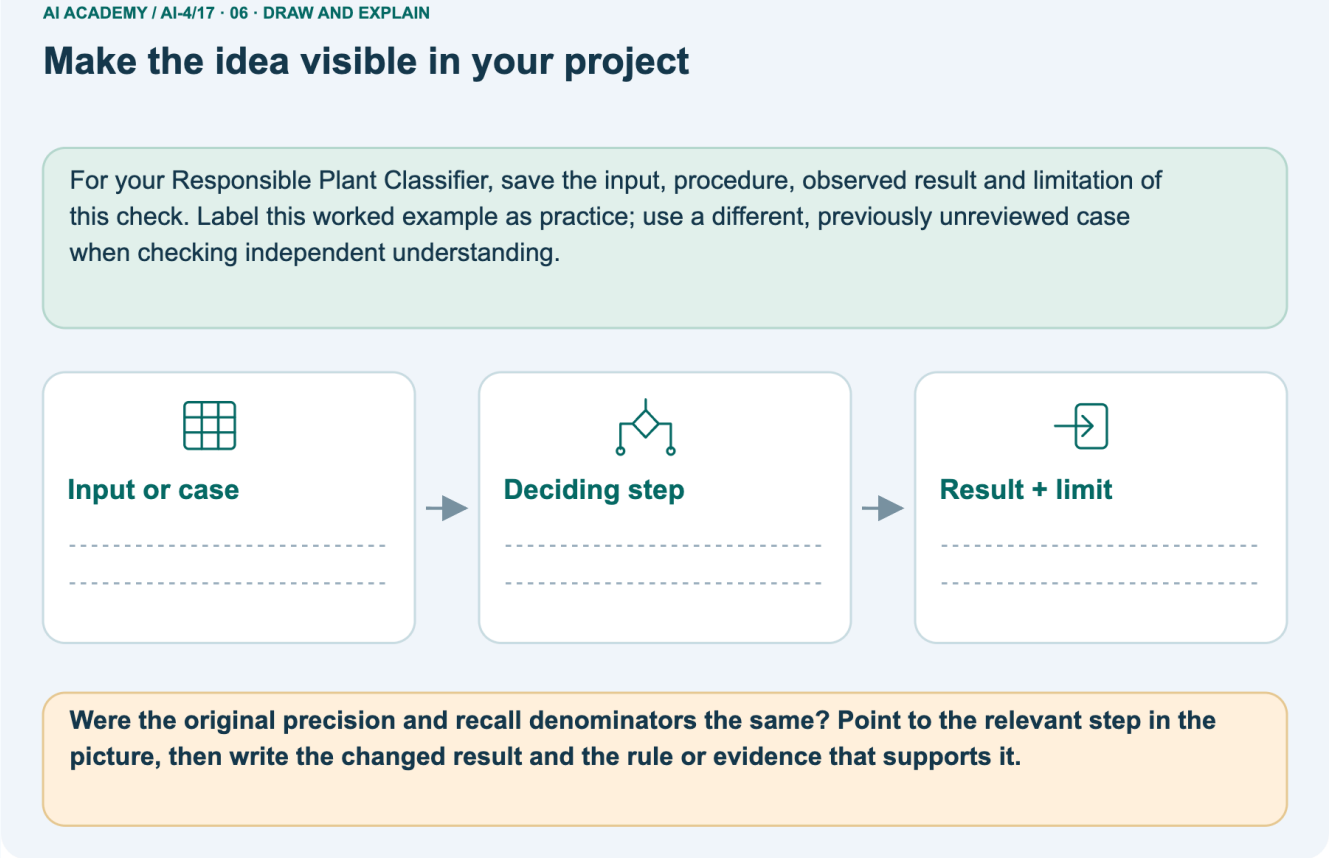
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. **Precision** used8 predicted-positive leaves; **recall** used10 actual-positive leaves. Their meanings remain distinct even when numbers later coincide.

Changed-case check: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.